

Multistate models

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Multistate models

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Multistate models

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ms-Markov

Common assumptions in survival analysis

1. Subjects are **either** “healthy” **or** “diseased”, with no intermediate state.
2. The disease is **irreversible**, or requires intervention to be cured.
3. The disease is **accurately** diagnosed.
4. The time of disease incidence is known **exactly**.

These assumptions are true for **death** and many **chronic diseases**.

A question of definition:

—we use occurrence of **recording of** a given disease

Accurate event dates?

- ▶ **Accurate dates of transition:**
complete observation of the process
all times and types of transitions between states are known
- ▶ **Panel data**
states known at certain times for each person
not necessarily the same times for different persons
- ▶ Panel data is analyzed with the **msm** package
this representation not treated further here

Markov models for multistate processes

- ▶ transition **probability** between two states ($A \rightarrow B$) depends only on current state through $t - s$:

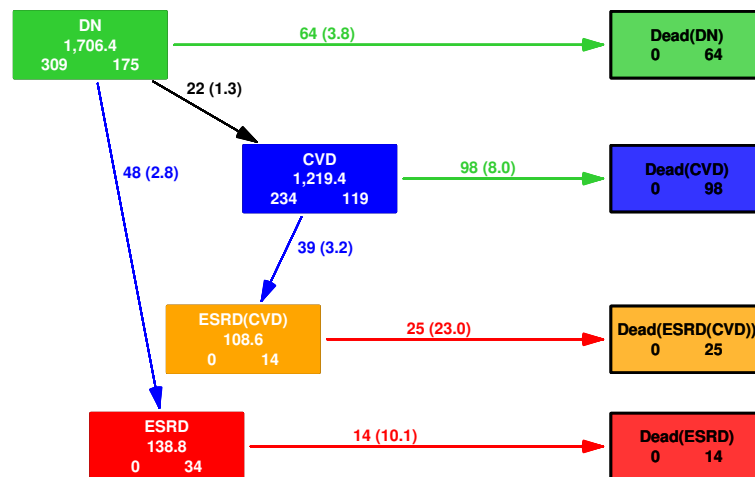
$$P \{ \text{state } B \text{ at } t \mid \text{state } A \text{ at } s \} = p_{AB}(t - s)$$

- ▶ — this is the **Markov** property
- ▶ $\Rightarrow$ transition **rates** are constant over time
- ▶ (time-fixed) covariates may influence transition rates
- ▶ the formal Markov property is **very** restrictive
- ▶ in the clinical literature, the term “Markov model” is often used about any type of multistate model

Components of a multistate (Markov) model

- ▶ Define the disease states
- ▶ Define which transitions between states are allowed
- ▶ Select covariates influencing transition rates (may be different between transitions)
- ▶ Not a trivial task — do we want e.g.
 - ▶ cause of death (CVD, Cancer, Other)
 - ▶ disease status at death (prev.CVD, prev.Can, both, neither)
- ▶ Types of parameters to target:
 - ▶ **Transition rates** between states
 - ▶ **Probability** of **state occupancy**
 - ▶ Expected **sojourn time** in a state

A more complicated multistate model



Multistate models (ms-Markov)

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Likelihood for a multistate model

- ▶ The likelihood of the model depends on the probability of being in state B at time t_1 , given that you were in state A at time t_0 .
- ▶ Assume transition rates constant in small time intervals, $\lambda^{A \rightarrow B}$
- ▶ $\Rightarrow$ each interval for a person contributes term(s) to the log-likelihood
- ▶ one term for each possible transition to a subsequent state
- ▶ the total log-likelihood for person p in state A during intervals i is a sum of these terms: $\ell_p = \sum_{i,B} d_{pi} (\log(\lambda_{pi}^{A \rightarrow B}) - \lambda_{pi}^{A \rightarrow B} y_{pi})$
- ▶ $\Rightarrow$ each term has the form of the likelihood for a Poisson variate d with mean λy : $d \log(\lambda) - \lambda y$

Multistate models (ms-Markov)

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Likelihood for a multistate model

- ▶ each term has the form of the likelihood for a Poisson variate d with mean λy
- ▶ terms are **not** independent, but the total likelihood is a **product**; hence of the same form as the likelihood from a set of independent (Poisson) variates
- ▶ but terms from intervals within one person are neither Poisson nor independent

Multistate models (ms-Markov)

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Realms of multistate modeling

- ▶ transition rates (intensities) — dimension time^{-1}
this is the scale of observation, (d, y) (complete data)
- ▶ state probabilities — dimensionless, time^0
 integral of intensities w.r.t. to time
- ▶ sojourn times — dimension time^1
 integral of state probabilities w.r.t. to time

Classes of multistate models

- ▶ Markov model: transition between states depends **only** on current state $\Rightarrow$ transition rates are constant
 $\text{time-homogeneous Markov model}$
- ▶ If transition rates depend on the **same timescale** only:
 $\text{time-inhomogeneous Markov model}$
- ▶ If transition rates depend on the time since entry to the current state we have a **semi-Markov model**
- ▶ If transition rates depend on several timescales we have a $\text{general multistate model}$ (there is no formal name for this)

...it is common-place in the literature to use the term “Markov model” for any type of multistate model.

Computing state probabilities from intensities in multistate models

- ▶ time-homogeneous Markov model:
closed-form formulae exist
- ▶ time-inhomogeneous Markov model:
closed-form formulae exist (a bit more complicated)
- ▶ **semi-Markov** model:
no closed form formulae exist
- ▶ general multistate model:
no closed form formulae exist

No formulae means that any inference on state probabilities and sojourn times must be based on (micro)**simulation** from the model.

Multistate models with Lexis

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ms-Lexis

Example: Renal failure data from Steno

Hovind P, Tarnow L, Rossing P, Carstensen B, and Parving H-H: Improved survival in patients obtaining remission of nephrotic range albuminuria in diabetic nephropathy. *Kidney Int.*, 66(3):1180–1186, 2004.

- ▶ Endpoint of interest: Death or end stage renal disease (ESRD), i.e. dialysis or kidney transplant.
- ▶ 96 patients entering at nephrotic range albuminuria (NRA), i.e. $U\text{-alb} > 300\text{mg/day}$.
- ▶ Is remission from this condition (i.e return to $U\text{-alb} < 300\text{mg/day}$) predictive of the prognosis?

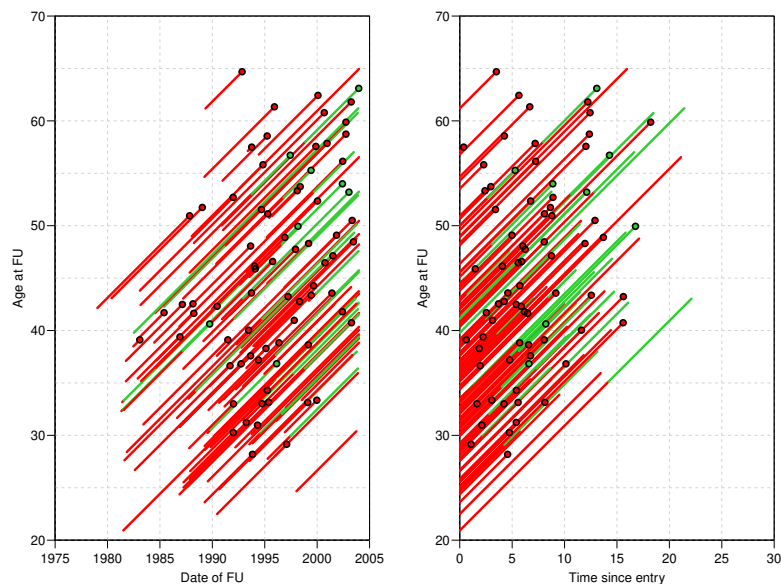
Multistate models with Lexis (ms-Lexis)

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	Total	Remission	
		Yes	No
No. patients	125	32	93
No. events	77	8	69
Follow-up time (years)	1084.7	259.9	824.8
Cox-model:			
Timescale: Time since nephrotic range albuminuria (NRA)			
Entry: 2.5 years of GFR-measurements after NRA			
Outcome: ESRD or Death			
Estimates:	RR	95% c.i.	p
Fixed covariates:			
Sex (F vs. M):	0.92	(0.53,1.57)	0.740
Age at NRA (per 10 years):	1.42	(1.08,1.87)	0.011
Time-dependent covariate:			
Obtained remission:	0.28	(0.13,0.59)	0.001

Multistate models with Lexis (ms-Lexis)

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Multistate models with Lexis (ms-Lexis)

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Features of the analysis

- ▶ Remission is included as a time-dependent variable.
- ▶ Age at entry is included as a fixed variable.

```
renal[1:5,]
id    dob      doe      dor      dox event
17 1967.944 1996.013      NA 1997.094     2
26 1959.306 1989.535 1989.814 1996.136     1
27 1962.014 1987.846      NA 1993.239     3
33 1950.747 1995.243 1995.717 2003.993     0
42 1961.296 1987.884 1996.650 2003.955     0
```

Note patient 26, 33 and 42 obtain remission.

Multistate models with Lexis (ms-Lexis)

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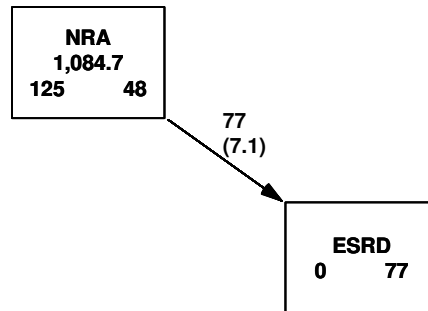
```
> Lr <- Lexis(entry = list(per = doe,
+                           age = doe - dob,
+                           tfi = 0),
+             exit = list(per = dox),
+             exit.status = event > 0,
+             states = c("NRA", "ESRD"),
+             data = renal)
> summary(Lr)

Transitions:
  To
From  NRA  ESRD  Records:  Events:  Risk time:  Persons:
  NRA   48   77      125      77    1084.67      125
```

Multistate models with Lexis (ms-Lexis)

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```
> boxes(Lr, boxpos = list(x = c(25, 75),
+                          y = c(75, 25)),
+       scale.R = 100, show.BE = TRUE )
```



Cutting follow-up at remission: `cutLexis`

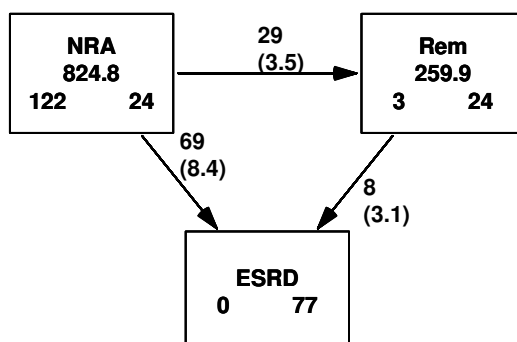
```
> Lc <- cutLexis(Lr, cut = Lr$dor,
+               timescale = "per",
+               new.state = "Rem")
> summary(Lc)
```

Transitions:

	From	To	NRA	Rem	ESRD	Records:	Events:	Risk time:	Persons:
NRA	24	29	69	122	98	824.77	122		
Rem	0	24	8	32	8	259.90	32		
Sum	24	53	77	154	106	1084.67	125		

Showing states and FU: `boxes.Lexis`

```
> boxes(Lc, boxpos = list(x = c(15, 85, 50),
+                          y = c(85, 85, 20)),
+       scale.R = 100, show.BE = TRUE)
```



Cutting follow up at events: `cutLexis`

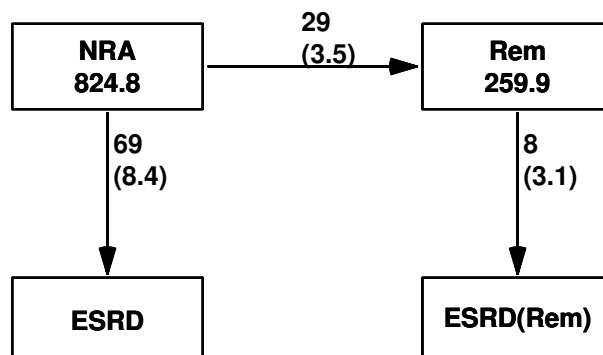
```
> Lc <- cutLexis( Lr, cut = Lr$dor,
+               timescale = "per",
+               new.state = "Rem",
+               split.states = TRUE)
> summary( Lc )
```

Transitions:

From	To	NRA	Rem	ESRD	ESRD(Rem)	Records:	Events:	Risk time:	Persons:
NRA	24	29	69	0	122	98	824.77	122	
Rem	0	24	0	8	32	8	259.90	32	
Sum	24	53	69	8	154	106	1084.67	125	

Showing states and FU: `boxes.Lexis`

```
> boxes(Lc, boxpos = list(x = c(15, 85, 15, 85),
+                          y = c(85, 85, 20, 20)),
+       scale.R = 100)
```



Likelihood for a general MS-model

- ▶ Product of likelihoods for each transition — each one as for a survival model
- ▶ **Risk time** is the risk time in the “From” state
- ▶ **Events** are transitions to the “To” state
- ▶ All other transitions out of “From” are treated as **censorings**
- ▶ Possible to fit models to `Lexis` objects
 - ▶ separately for each transition
 - ▶ jointly for transitions **from different states to the same state**
 - ▶ jointly for transitions **from the same state to different states**: normally not meaningful, so **don't!**

Calculating state probabilities

$$\begin{aligned} P \{ \text{Remission \textbf{before} time } t \} \\ &= \int_0^t \lambda(u) \exp \left(- \int_0^u \lambda(s) + \mu_{\text{NRA}}(s) \, ds \right) \, du \\ \\ P \{ \text{Being in remission \textbf{at} time } t \} \\ &= \int_0^t \lambda(u) \exp \left(- \int_0^u \lambda(s) + \mu_{\text{NRA}}(s) \, ds \right) \times \\ &\quad \exp \left(- \int_u^t \mu_{\text{rem}}(s) \, ds \right) \, du \end{aligned}$$

Note μ_{rem} could also depend on u , time since obtained remission.

Multistate models with `Lexis` (ms-Lexis)

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Sketch of programming, assuming that λ (`lambda`), μ_{NRA} (`mu.nra`) and μ_{rem} (`mu.rem`) are known at any age (stored in vectors)

```
c.rem      <- cumsum(lambda)
c.mort.nra <- cumsum(mu.nra)
c.mort.rem <- cumsum(mu.rem)
pr1 <- cumsum(lambda * exp(-(c.rem + c.mort.nra)))

integr(t,s) <-
function(t,s){
  lambda[s] * exp(-(c.rem[s] + c.mort.nra[s])) *
    exp(-(c.mort.rem[t] - c.mort.rem[s]))
}
for(t in 1:100) p2[t] <- sum(integr(t,1:t))
```

If μ_{rem} also depends on time since remission, then `c.mort.rem` should have an extra argument—technically very complicated

Analytical approach is not recommended — simulation is simpler

Multistate models with `Lexis` (ms-Lexis)

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Prediction in multistate models: `simLexis` and renal failure

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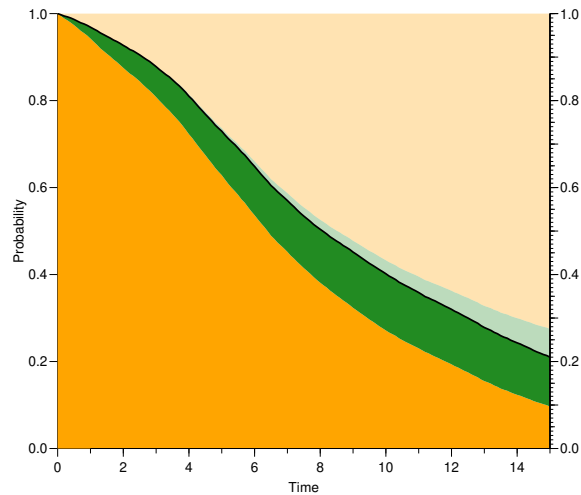
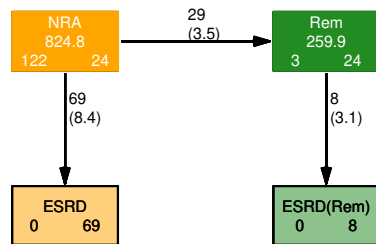
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simRenal



Modeling rates in a multistate model

Each transition modeled by a model for rates
 Cox-model or Poisson-model for split data: `glm` or `gam`
 —requires that follow-up is split in small intervals:

```
> sLc <- splitLexis(Lc, "tfi", breaks = seq(0, 30, 1/12))
> summary(sLc, t = T)
```

Transitions:

From	To	NRA	Rem	ESRD	ESRD(Rem)	Records	Events	Risk time	Persons
NRA	NRA	9854	29	69	0	9952	98	824.77	122
Rem	Rem	0	3139	0	8	3147	8	259.90	32
Sum		9854	3168	69	8	13099	106	1084.67	125

Timescales:

```
per age tfi
" " " " " "
```

Modeling rates in a multistate model

```
> # Remission-rate
> mr <- gam(cbind(lex.Xst == "Rem", lex.dur)
+ ~ s(tfi, k = 10) + sex,
+ family = poisreg,
+ data = subset(sLc, lex.Cst == "NRA"))
> #
> mr <- gamLexis(sLc,
+ ~ s(tfi, k = 10) + sex,
+ to = "Rem" )

mgcv::gam Poisson analysis of Lexis object sLc with log link:
Rates for the transition:
NRA->Rem
```

... using the Lexis properties

```
> # ESRD-rates
> mx <- gam(cbind(lex.Xst %in% c("ESRD", "ESRD(Rem)"), lex.dur)
+ ~ s(tfi, k = 10) + sex + I((doe - dob - 40) / 10) +
+ I(lex.Cst == "Rem"),
+ family = poisreg,
+ data = subset(sLc, lex.Cst %in% c("NRA", "Rem")))
> #
> mx <- gamLexis(sLc,
+ ~ s(tfi, k = 10) + sex + I((doe - dob - 40) / 10) +
+ lex.Cst )

mgcv::gam Poisson analysis of Lexis object sLc with log link:
Rates for transitions:
NRA->ESRD
Rem->ESRD(Rem)
```

Default is to model all transitions **to** absorbing states

State probabilities

How do we get from rate-models (and origin) to state probabilities:

- 1 Analytic calculations:
 - ▶ immensely complicated formulae
 - ▶ computationally fast (once implemented)
 - ▶ difficult to generalize
- 2 Simulation of persons' histories
 - ▶ conceptually simple
 - ▶ computationally not quite simple
 - ▶ easy to generalize
 - ▶ hard to get confidence intervals (bootstrap)

Simulation of a survival time

- ▶ For a rate function $\lambda(t)$, $\Lambda(t) = \int_0^t \lambda(s) ds$:

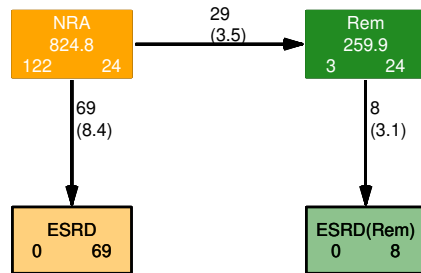
$$S(t) = \exp(-\Lambda(t))$$

- ▶ Simulate a survival probability, uniform $u \in [0, 1]$:

$$u = S(t) \Leftrightarrow \Lambda(t) = -\log(u)$$

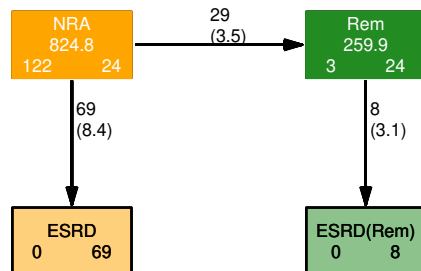
- ▶ Knowledge of $\Lambda(t)$ makes it easy to find a survival time t
 - essentially just linear interpolation
- ▶ $\Lambda(t)$ is just a vector of values at large set of times

Simulation in a multistate model



- ▶ Simulate a “survival time” for each transition **out** of a state.
- ▶ The smallest of these is the transition time.
- ▶ Choose the corresponding transition type as transition.

Transition objects are fitted `coxph/glm/gam` models



```

> Tr <- list("NRA" = list("ESRD" = mx,
+                          "Rem" = mx),
+            "Rem" = list("ESRD(Rem)" = mx))
  
```

`simLexis`

Input required:

- ▶ A `Lexis` object with the initial state of the persons to be simulated.
(`lex.dur` and `lex.Xst` will be ignored—they are outcomes to be simulated)
- ▶ A transition object with the estimated Poisson models collected in a list of lists.

Output produced:

- ▶ A `Lexis` object with simulated event histories for many persons

Using `simLexis` I

A new `Lexis` object (`init`, say), where each row represents a person with desired covariate values.

`init` must have same structure as the one used for estimation — time scales must be initiated even if not used in models

```
> init <- sLc[NULL, c(timeScales(sLc), "lex.Cst")]
> init[1,"per"] <- 1994
> init[1,"age"] <- 40
> init[1,"tft"] <- 0
> init[1,"lex.Cst"] <- "NRA"
> init[1,"sex"] <- "M"
> init[1,"dob"] <- 1954
> init[1,"doe"] <- 1994
> init

      per age tft lex.Cst sex  dob  doe
1994  40   0    NRA    M 1954 1994
```

Using `simLexis` II

```
> system.time( sim1 <- simLexis(Tr, init, N = 10000, t.range = 15.1) )
      bruger      system forløbet
      21.16      2.53      23.95

> summary(sim1)
Transitions:
      To
From  NRA  Rem ESRD ESRD(Rem) Records: Events: Risk time: Persons:
NRA  942 1787 7271          0    10000    9058  71993.67    10000
Rem   0 1133   0          654    1787     654  15138.13     1787
Sum  942 2920 7271          654   11787    9712  87131.80    10000
```

This is a simulated cohort of 10,000 persons with NRA, aged 40 in 1994.

Using a simulated `Lexis` object — `pState` I

```
> NN <- nState(sim1, at = seq(0, 15, 0.1),
+             from = 0,
+             time.scale = "tft")
> head(NN)

      State
when  NRA  Rem ESRD ESRD(Rem)
0    10000   0   0       0
0.1  9944   29  27       0
0.2  9888   56  56       0
0.3  9844   80  76       0
0.4  9792  106 102       0
0.5  9735  130 134       1

> sP1 <- pState(NN, perm = c(1, 2, 4, 3))
> head(sP1, 3)
```

Using a simulated Lexis object — pState II

```
      State
when   NRA    Rem ESRD(Rem) ESRD
0      1.0000 1.0000    1.0000    1
0.1    0.9944 0.9973    0.9973    1
0.2    0.9888 0.9944    0.9944    1

> tail(sP1, 3)

      State
when   NRA    Rem ESRD(Rem) ESRD
14.8  0.1019 0.2161    0.2799    1
14.9  0.0997 0.2138    0.2781    1
15     0.0965 0.2100    0.2752    1
```

Using a simulated Lexis object — pState III

```
> par(mar = c(3, 3, 0.5, 2), mgp = c(3, 1, 0) / 1.6, las = 1)
> plot(sP1, col = clr[c(1, 2, 3, 4)], xlim = c(0,15) )
> lines(as.numeric(rownames(sP1)), sP1[,2], lwd = 2)
> axis(side = 1, at = 1:55, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:5 / 5)
> axis(side = 4, at = 0:10 / 10, labels = NA)
> axis(side = 4, at = 0:20 / 20, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:100/100, labels = NA, tcl = -0.2)

> sP2 <- pState(NN, perm = c(4, 2, 1, 3))
> head(sP2, 3)

      State
when ESRD(Rem)    Rem    NRA ESRD
0      0 0.0000 1.0000    1
0.1      0 0.0029 0.9973    1
0.2      0 0.0056 0.9944    1
```

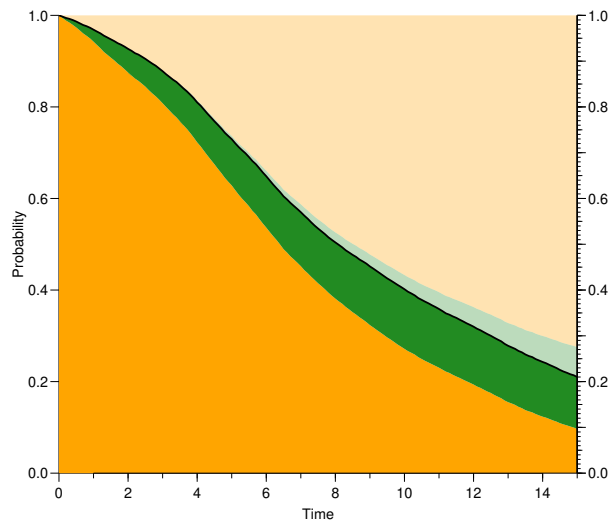
Using a simulated Lexis object — pState IV

```
> tail(sP2, 3)

      State
when ESRD(Rem)    Rem    NRA ESRD
14.8  0.0638 0.1780 0.2799    1
14.9  0.0643 0.1784 0.2781    1
15     0.0652 0.1787 0.2752    1

> par(mar = c(3, 3, 0.5, 2), mgp = c(3, 1, 0) / 1.6, las = 1)
> plot(sP2, col = clr[c(3, 2, 1, 4)])
> axis(side = 1, at = 1:55, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:5 / 5)
> axis(side = 4, at = 0:10 / 10, labels = NA)
> axis(side = 4, at = 0:20 / 20, labels = NA, tcl = -0.3)
> axis(side = 4, at = 0:100/100, labels = NA, tcl = -0.2)
```

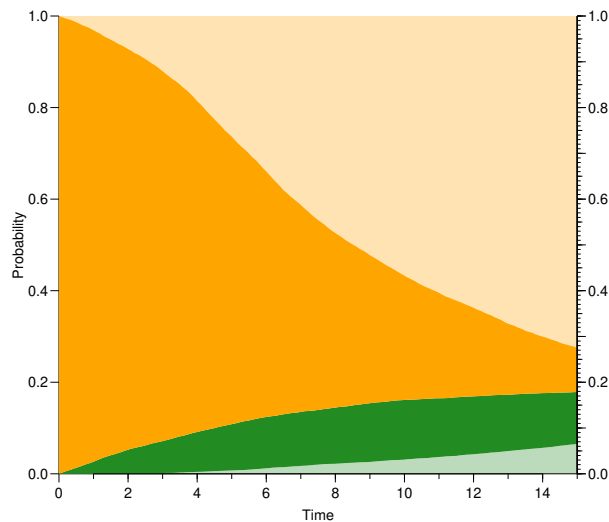
Simulated probabilities



Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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Simulated probabilities



Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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How many persons should you simulate?

- ▶ All probabilities have the same denominator — the initial number of persons in the simulation, N , say.
- ▶ Thus, any probability will be of the form $p = x/N$
- ▶ For small probabilities we have that:

$$\text{s.e.}(\log(\hat{p})) = (1 - p)/\sqrt{Np(1 - p)}$$

- ▶ So c.i. has the form $p \times \text{erf}$ where:

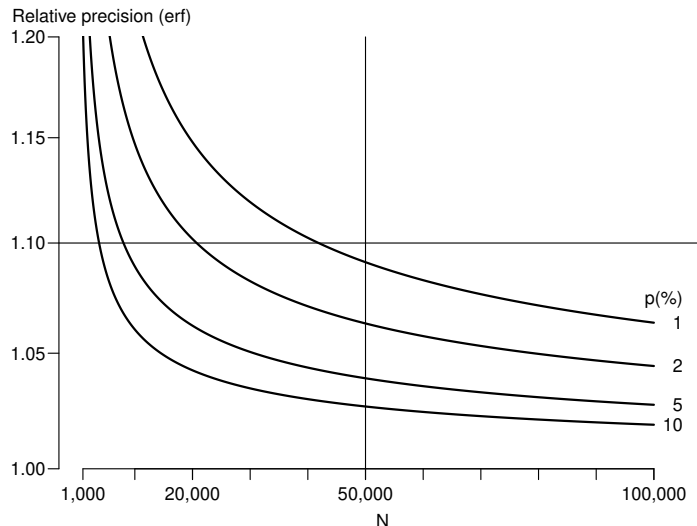
$$\text{erf} = \exp(1.96 \times (1 - p)/\sqrt{Np(1 - p)})$$

- ▶ 40,000

Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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Precision of simulated probabilities



Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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Using a simulated `Lexis` object — sojourn time I

```
> summary(sim1)
Transitions:
  To
From  NRA  Rem  ESRD  ESRD(Rem)  Records  Events  Risk time  Persons:
  NRA  942 1787 7271           0    10000    9058  71993.67    10000
  Rem   0 1133   0         654     1787     654  15138.13     1787
  Sum  942 2920 7271         654    11787    9712  87131.80    10000

> xtabs(lex.dur / 10000 ~ lex.Cst, data = sim1) |> addmargins()
lex.Cst
      NRA      Rem      ESRD  ESRD(Rem)      Sum
7.199367 1.513813 0.000000 0.000000 8.713180
```

During the first 15 years after entry, the expected time in remission is 1.5 years out of an expected lifetime of 8.8 years, for a 40 year old man

Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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Multistate model overview

- ▶ Clarifies the relevant states and transitions
- ▶ Allows proper estimation of transition rates
- ▶ — and relationships between them
- ▶ Separate model for each transition
- ▶ The usual survival methodology to compute probabilities breaks down
- ▶ Using transition rates for simulation allows estimation of:
 - ▶ transition rates (as usual)
 - ▶ cumulative probabilities (**not** quite as usual)
 - ▶ sojourn times (how long in each state)

Prediction in multistate models: `simLexis` and renal failure (`simRenal`)

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